

# Yiquan Wang

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• Member of IEEE Biometrics Council, Chinese Chemical Society & Biophysical Society of China

## Education

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>National Base for Mathematical Research and Teaching Talents, Xinjiang University</b><br><i>B.S. in Mathematics and Applied Mathematics</i>                                                  | Urumqi, Xinjiang<br>2023.09 – 2027.06    |
| <b>Tsien Excellence in Engineering Program, Tsinghua University &amp; Shenzhen X-Institute</b><br><i>Joint Program, X Scholar</i> — Representation learning for protein function and dynamics   | Shenzhen, Guangdong<br>2024.06 – 2027.06 |
| <b>Institute of Neurological and Psychiatric Disorders, Shenzhen Bay Laboratory</b><br><i>Visiting Student, Wen Yuan's Research Group</i> — Single-cell bioinformatics for neurobiology studies | Shenzhen, Guangdong<br>2025.07 – 2025.09 |
| <b>Institute of Systems and Physical Biology, Shenzhen Bay Laboratory</b><br><i>Visiting Student, Kai Huang's Research Group</i> — IDP, LLPS & genome folding, undruggable-target design        | Shenzhen, Guangdong<br>Starting 2026.07  |

## Selected Publications

Name in bold; † co-first author; \* corresponding author.

### Direction 1: Representation Learning & Biomolecular Mechanisms

- [1] **Wang, Y.**, et al. (2025). From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction. *Journal of Chemical Information and Modeling*.
- [2] **Wang, Y.\***, et al. (2026). Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors. *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026*.
- [3] **Wang, Y.\***, et al. (2026). Integrative transcriptomics, machine learning, and molecular docking derive a DAM-like macrophage signature for risk stratification and therapeutic nomination in glioblastoma. *Computational Biology and Chemistry*.
- [4] **Wang, Y.†**, et al. (2025). Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules versus Therapeutic Peptides. *Biology*.
- [5] Wang, X.†\*, **Wang, Y.†\***, et al. (2025). Octopus Inspired Optimization (OIO): A Hierarchical Framework for Navigating Protein Fitness Landscapes. In *2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE.
- [6] **Wang, Y.\***, et al. (2026). Comment on: “GLM7 – A Novel Composite Glycolipid Index Derived from Routine Health Indicators for Enhanced Diagnosis and Prediction of Multimorbidity”. *Advanced Science*. (correspondence).
- [7] **Wang, Y.\***, et al. (2025). AI for disease prediction: Performance insights and key limitations. *Journal of Clinical Neuroscience*. (correspondence).

### Direction 2: First-Principles of Biological Complexity Based on Mathematical Physics

- [8] **Wang, Y.\*** (2026). Sub-residue sharpness of protein helix-coil transitions reveals a spatial-spectral uncertainty limit. *arXiv preprint arXiv:2602.21787*.
- [9] **Wang, Y.\*** (2026). Piecewise integrability of the discrete Hasimoto map for analytic prediction and design of helical peptides. *arXiv preprint arXiv:2602.16255*.
- [10] **Wang, Y.\*** (2026). Structural barriers of the discrete Hasimoto map applied to protein backbone geometry. *arXiv preprint arXiv:2602.13160*.

### Direction 3: Interdisciplinary Explorations & Collaborative Synergies

- [11] **Wang, Y.**, Zai, J., Liu, Z., Chen, J., & Sabir, E.\* (2026). Hamiltonian Connectivity in  $k$ -Ary  $n$ -Cubes under a Region-Based Fault Model. *Information Sciences*.
- [12] **Wang, Y.\***, et al. (2026). AI Driven Discovery of Bio Ecological Mediation in Cascading Heatwave Risks. *Physics and Chemistry of the Earth*.
- [13] **Wang, Y.\***, et al. (2026). Geometric Fragility in Alzheimer's Disease: Probing the Loss of Hippocampal Hierarchical Abstraction via Contrastive Point Cloud Modeling. In *2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE.
- [14] Cai, M., ..., **Wang, Y.\***, & Wei, K.\* (2025). APOE  $\epsilon$ 4-Associated Hippocampal Atrophy Trajectories Across the Alzheimer's Disease Continuum: A Systematic Review, Meta-Analysis, and Longitudinal Validation. *Frontiers in Aging Neuroscience*.

## Research Projects

- Tsinghua University Tsien Excellence in Engineering Program ESRT** 2024.08-2025.10
- **Project:** From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction.
  - Motivated by an intuition that protein structure mirrors music—catalytic sites resembling melodic climaxes and  $\beta$ -sheets resembling calmer passages—I encoded amino acid sequences as 2D spectrograms (protein sonification) for function prediction. Ablations across an 18,000-sequence/12-class benchmark showed the 1D→2D representational structure itself, rather than explicit biophysical priors, was the key driver of performance, matching ESM-2 and ProtBERT and reaching 90.44% on the external CARE benchmark.
  - [Project Repository](#). Supervised by Prof. [Kai Wei](#).

- Chinese Academy of Sciences (CAS) Innovation Practice Training Program** 2024.11 – 2025.09

- **Project:** A Knowledge Graph-based Q&A System for Heatwave Disaster Adaptation.
- Built HeDA, an autonomous synthesis agent that constructs a knowledge graph (34,933 entities, 42,890 relations) from 8,365 publications to map compound-heatwave cascading risks. On a temporally held-out multi-hop benchmark, graph augmentation improved accuracy by

- 8.1–9.8% over standalone foundation models; topological analysis surfaced a bio-ecological mediation effect, where biological systems act as the primary structural mediators transforming physical hazards into systemic socioeconomic losses.
- [Project Repository](#). Supervised by Researcher [Yong Ge](#).

**Provincial Undergraduate Innovation Training Program**

2024.03 – 2025.06

- **Project:** Research on the Generation Cyclability of Cartesian Product Graphs and Related Problems.
- Studied Hamiltonicity of the  $k$ -ary  $n$ -cube  $Q_n^k$ , a classical graph-theoretic interconnection topology. Motivated by the fact that real-world faults are spatially clustered rather than independent, I introduced the Region-Based Fault (RBF) model characterizing faults as connected subgraphs, and gave a constructive proof that  $Q_n^k$  remains Hamiltonian-connected for odd  $k \geq 3$ ,  $n \geq 2$  under RBF conditions, via an adaptive graph-decomposition argument robust well beyond its theoretical guarantees.
- [Project Repository](#). Supervised by Prof. [Eminjan Sabir](#).

**National Undergraduate Innovation Program**

2025.05 – 2026.12

- **Project:** Conditional Diffusion Model on Copy Number Variation for Alzheimer's Disease Risk Assessment.

**Tsinghua University Tsien Excellence in Engineering Program ORIC**

2025.12 – 2026.12

- **Project:** Application of Artificial Intelligence in Whole Genome Selection Breeding.

**Chinese Academy of Sciences (CAS) Innovation Practice Training Program**

2025.11 – 2026.11

- **Project:** Urban Big Data, Multimodal Fusion and Spatial Intelligence Analysis (Second Author).

**Professional Experience**

**Beijing Frontier Research Center for Biological Structure, Tsinghua University**

Beijing, China  
Intern
2025.07 – Present

- Developed the [Peptide Design Competition Website](#), the [Polypeptide Structure Database](#) and engaged in protein and polypeptide design research. Supervised by Prof. Yafei Yuan and [Prof. Tong Wang](#).

**Institute of Software, CAS & Huawei Mindspore Community**

Remote  
Intern
2024.09 – 2025.03

- Implemented a VGG19-based model for Pollock-style art generation, focusing on fractal and turbulent feature extraction and the creation of NFT-based digital labels. Featured on Huawei's official Wechat.

**Awards & Honors**

- **China Undergraduate Life Science Contest:** National Second Prize 2026
- **“Challenge Cup” China College Students’ Extracurricular Academic Science and Technology Works Competition:** Autonomous Region Second Prize 2026
- **China College Students’ E-Commerce “Innovation, Creativity and Entrepreneurship” Challenge:** Autonomous Region Second Prize 2026
- **China College Student Self-improvement Star** 2025
- **UN Global Leadership and ESG Programme × X-Institute Shenzhen "AI for the Young and the Elderly" Challenge:** Best Innovation Award 2026
- **Smart Chemical City Challenge (Xinjiang Station):** Second Prize 2025
- **Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM):** National Second Prize 2025
- **SynBio Challenges:** Silver Award × 2 2025
- **2025 X-Fusion "Global Innovator Fusion Conference":** Best Poster Award 2025
- **The Mathematical Contest in Modeling (MCM):** Honorable Mention 2025
- **Alibaba Cloud Tianchi University Student Competition:** National Finals, 17th Place 2024
- **APMCM Asia-Pacific Mathematical Modeling Competition:** National Third Prize 2024

**Academic Activities**

**Peer Reviewer:** ICLR/ICML/NeurIPS workshops; Urban Climate; Physics and Chemistry of the Earth; Mini-Reviews in Medicinal Chemistry; Current Science; F1000Research.

**Academic Engagement:**

- Tsinghua University-Peking University Center for Life Sciences Summer Camp 2025.07
- Shenzhen Bay Laboratory / Shenzhen Medical Academy of Research and Translation Summer Research 2025.07 – 2025.09
- Tsinghua University Tsien Excellence in Engineering Program – Zero One Scholar 2024.06 – 2027.06
- AI Winter School, Brown University Department of Physics 2025.01
- CAAI Artificial Intelligence and Technology Ethics Training Course 2024.09 – 2024.12
- Fudan University Summer School of Mathematical Logic 2024.08
- Wuhan University National Tianyuan Mathematics Center Discussion Class 2024.03 – 2024.06

**Website Development**

- **Peptide Design Competition, Tsinghua University** — [\[Link\]](#)
- **Polypeptide Structure Database, Tsinghua University** — [\[Link\]](#)
- **Tong Wang (Tsinghua University) Research Group** — [\[Link\]](#)